

Mohammed Alaklounk

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Location: Gebze/Turkey

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Education

Gebze Technical University
BSc Computer Engineering
Sep 2024 – Jul 2028 (expected) · GPA 3.55

Projects

Scree

A falling-sand physics simulator in C++20 and raylib where every pixel is an independent particle which reacts with the materials around it. It runs in the browser via WebAssembly.

- 30+ materials that spread, burn and dissolve into one another through a tag-based reaction system
- Materials, reactions and physics are defined in editable JSON, and everything can be hot reloaded
- Users can create, edit and delete materials in-app
- Users can import and export your own materials and canvases
- Bloom lighting for glowing materials

Tech stack: C++20, raylib, Dear ImGui, nlohmann/json, CMake, Catch2, Emscripten/WASM

[Live Demo](#)

Latch

A logic-gate sandbox and simulator in C++20, raylib and Dear ImGui. You can place gates, wire them together and simulate the circuit with one-tick propagation delay per gate.

- Three logic values: LOW, HIGH and UNDEFINED for floating inputs
- Nine basic gates: AND, OR, XOR and their complements, NOT, a toggle switch and an LED
- Wires are graphs of nodes and segments which react to node and segment deletion
- Wires auto-route when extended and reroute when a gate or group of gates is moved
- Build multiple circuits in the same instance through sheets with a shared component palette and clipboard
- Canvas selection, copy/paste, delete, and full undo/redo
- A headless simulation core inside its own static library tested with Catch2

Tech stack: C++20, raylib, Dear ImGui, CMake, Catch2

Skills

- C++
- Python
- Dear ImGui
- Catch2
- Emscripten (C++ to web)
- raylib (Graphics)
- CMake
- Git
- OpenGL

Languages

Arabic (Native) · English (Fluent) · Turkish (B1)